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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	T.S.P Madhuri
Student Name	V.PRANAV
Roll Number	2520030207
Branch	CSE
Course Outcome	CO-5

Case Study Topic: Spotify Monthly-Listener Leaderboard — Top K Artists using Min-Heap

Work Done Summary:

In this case study, we implemented a Min-Heap in Java to simulate Spotify's monthly-listener leaderboard system. The Min-Heap maintained the top K artists with the highest listener counts while scanning artist records. This approach efficiently identified the most-streamed artists without sorting the entire dataset and reduced computational cost for large-scale analytics.

Implementation Details:

1. Created an Artist class to store artist names and monthly listener counts.
2. Implemented a Min-Heap using Java PriorityQueue.
3. Inserted the first K artists into the heap.
4. Compared each new artist's listener count with the heap root.
5. Replaced the root whenever a larger listener count was found.
6. Maintained heap size at K throughout processing.
7. Extracted and displayed the final Top-K artists sorted by listener count.

Result: Screenshots / Output

```
PS C:\Users\duddu\OneDrive\Desktop\rohith\CASE STUDY 5> java Main
Spotify Top 5 Artists

Artist10 -> 91M listeners
Artist6 -> 89M listeners
Artist3 -> 78M listeners
Artist8 -> 67M listeners
Artist5 -> 56M listeners

Time Complexity: O(n log k)
```

The Min-Heap successfully identified the Top 5 artists with the highest monthly listener counts.
Input Listener Counts: 45, 12, 78, 23, 56, 89, 34, 67, 18, 91, 50, 39
Top 5 Artists:
Artist10 - 91M
Artist6 - 89M
Artist3 - 78M
Artist8 - 67M
Artist5 - 56M
Time Complexity:
- Min-Heap Insertion: O(log k)
- Top-K Processing: O(n log k)

GitHub Link: https://github.com/pranavvv-18/DSA-2-CASE-STUDY

Student Signature	Faculty Signature
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